RITHICK RAMAN

Full Stack Developer | React.js | TypeScript | Node.js | PostgreSQL | AI/LLM Integration

Nilgiris, Tamil Nadu, India | +91 96267 59001 | rithickraman855@gmail.com | linkedin.com/in/rithick-r-819b72217

PROFESSIONAL SUMMARY

Full Stack Developer with 3+ years of experience owning features end-to-end, from database and API design to polished, responsive UI, in a fast-paced environment. Strong in JavaScript/TypeScript, React.js, Node.js, Express.js, FastAPI, and PostgreSQL, with hands-on experience in REST API design, state management, JWT/RBAC security, real-time systems (Socket.IO), Docker, and CI/CD (GitHub Actions, Jenkins). Built AI-powered document automation pipelines using Tesseract OCR and locally hosted LLMs (Ollama) for enterprise invoice and workflow processing. Delivered measurable results across production systems, including a 49% API response-time reduction, 40% faster frontend data loading, and 85% Jest test coverage.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, SQL, Python

Frontend: React.js, Redux, Context API, HTML5, CSS3, Responsive Design, Material UI, Tailwind CSS, Bootstrap

Backend & APIs: Node.js, Express.js, FastAPI, RESTful APIs, Swagger, Postman

Databases: PostgreSQL, MySQL, Prisma ORM, Redis

AI & Automation: Ollama (Local LLMs), LLM Integration, Tesseract OCR, IMAP Email Ingestion, Azure AI Pipelines

Auth & Security: JWT, OAuth2, RBAC, Session Management, API Security

Real-Time & Async: Socket.IO, BullMQ, Redis, PostgreSQL LISTEN/NOTIFY, Event-Driven Architecture

DevOps & Testing: Docker, GitHub Actions, Jenkins, Git, GitHub, CI/CD Pipelines, Jest (85% coverage), Agile/Scrum

PROFESSIONAL EXPERIENCE

Software Developer (Full Stack) — Adeptview, Chennai **August 2024 – Present**

- Designed, built, and deployed 3 production full-stack applications using React.js, TypeScript, Node.js, and PostgreSQL, owning features from database schema and API design to responsive UI and reducing time-to-market by 20% through modular component architecture and reusable API layers.
- Engineered RESTful APIs with targeted SQL query optimization and database indexing, reducing average API response time from 350 ms to 180 ms (49% improvement).
- Redesigned data-intensive dashboards with server-side pagination, multi-column filtering, and virtualized rendering, improving frontend data load performance by 40% on large datasets.
- Automated enterprise document workflows by integrating Tesseract OCR, LLM-based field extraction with locally hosted Ollama models, and Azure event-driven pipelines, significantly reducing manual invoice processing effort.
- Implemented JWT-based authentication and Role-Based Access Control (RBAC) for 5 user roles, establishing a secure, auditable authorization layer across production applications.
- Developed real-time notification infrastructure using Socket.IO and PostgreSQL LISTEN/NOTIFY, enabling event-driven UI updates with sub-second latency for connected clients.
- Implemented asynchronous job queues using BullMQ and Redis, decoupling time-intensive processing from the request cycle and improving system throughput and resilience.
- Set up CI/CD pipelines with GitHub Actions and Jenkins and containerized services with Docker to automate builds, testing, and deployments.
- Collaborated with product, design, and backend teams in an Agile/Scrum environment to translate requirements into working software.

Junior Software Developer — Adeptview, Chennai **August 2023 – August 2024**

- Developed and maintained RESTful APIs backed by PostgreSQL and Prisma ORM, ensuring type-safe database access, clean schema migrations, and reliable data layer contracts.
- Built and modernized responsive UI components using React.js and Material UI; integrated the frontend with Node.js/Express.js REST APIs, improving UI consistency and reducing integration defects.
- Led Jest-based automated testing initiatives, achieving 85% code coverage and reducing production defects by 25%.
- Improved database query performance by 20–30% through SQL optimization, index tuning, and PostgreSQL execution plan analysis.
- Integrated third-party APIs and designed automated data processing workflows, reducing manual effort in recurring data synchronization tasks.

KEY PROJECTS

Invoice Processing & Automation System (Full-Stack | AI/LLM | OCR | Async | Real-Time)

- Built an end-to-end document automation platform that ingests invoices via IMAP email (ImapFlow), extracts structured data using Tesseract OCR and LLM-based field recognition powered by locally hosted Ollama models, and routes documents through configurable approval workflows.
- Designed async processing pipelines with BullMQ and Redis, handling high-volume document queues independently of the main request cycle to ensure reliability under load.
- Implemented RBAC access control and real-time status notifications via Socket.IO, enabling multi-role team visibility across document lifecycle stages.
- Exposed a Python FastAPI microservice for AI/OCR processing, integrated with the Node.js backend via REST and documented with Swagger; automated build and test workflows with GitHub Actions.

Stack: React.js, TypeScript, Node.js, Express.js, PostgreSQL, Prisma ORM, BullMQ, Redis, Tesseract OCR, Ollama, FastAPI, ImapFlow, Socket.IO, JWT, Swagger, GitHub Actions

Inventory & Point-of-Sale (POS) Management System (Full-Stack | Microservices | Real-Time)

- Developed a full-featured inventory and order management system with Purchase Order (PO), Sales Order (SO), GRN, and GIN modules; implemented multi-location stock tracking and supplier management.
- Architected a Docker-based microservices deployment isolating billing, inventory, and notification services for independent scalability; integrated real-time stock updates via Socket.IO.

Stack: React.js, Node.js, Express.js, PostgreSQL, Prisma ORM, Docker, Socket.IO, JWT

Contract & Invoice Workflow Automation System (Full-Stack | Event-Driven | RBAC)

- Built a contract lifecycle management system with version control, audit trail logging, and configurable multi-stage approval workflows; implemented BullMQ + Redis event-driven job processing for async notifications and background tasks.
- Developed multi-organization data isolation with granular RBAC permissions and integrated ReactQuill for structured contract authoring with revision history.

Stack: React.js, Redux, Context API, Node.js, Express.js, PostgreSQL, Prisma ORM, BullMQ, Redis, ReactQuill, JWT

CERTIFICATIONS

Node.js (Basic & Intermediate) — HackerRank **July 2024**

Power BI Fundamentals (PL-100) — Microsoft **April 2024**

EDUCATION

B.E., Computer Science and Engineering **June 2019 – June 2023**

Bannari Amman Institute of Technology | CGPA: 8.0 / 10 (80%)